

What a prediction model figures out about its test questions
and the one it learns only roughly

Wenrui Yuan

The setup

We train a model to do one job, predict whether a learner answers the next question correctly. Answers can be right or wrong, or graded into a few levels (right or wrong is two levels, a five point grade is five).

To predict well, the model ends up working out three numbers on its own.

- **score**, how strong the learner is.
- **difficulty**, how hard the question is.
- **sharpness**, how clearly a question tells apart strong and weak learners.

On a sharp question, strong learners almost always get it and weak ones almost never. On a dull question, everyone is near a coin flip no matter their strength.

We never hand the model these numbers. It is graded only on predicting answers. On made-up data the true numbers are known, so we can check how well the model's numbers match them, from 0 (no match) to 1 (perfect order).

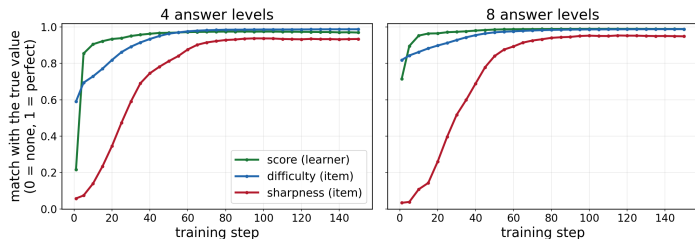
The question

When a model learns only by predicting answers, which of the three numbers does it get right, how fast, and why?

Short answer. It gets the learner's score and the question's difficulty easily. It struggles with the question's sharpness. The rest of the talk is why, what helps, and one tempting explanation we tested and ruled out.

Two of the three numbers are easy, one is not

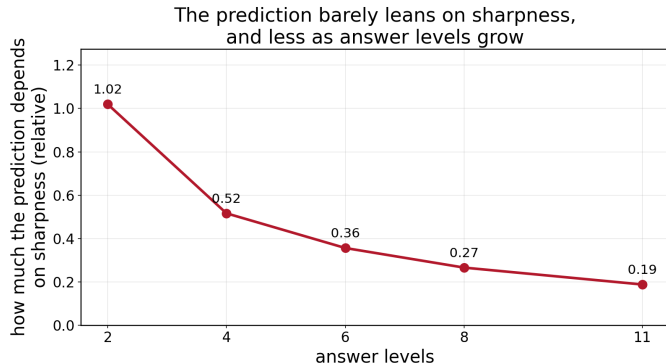
Score and difficulty are learned fast; sharpness lags



Score and difficulty climb to a near perfect match within the first few to twenty training steps. Sharpness rises slowly and never quite catches up, settling a notch below the other two (about 0.93 on the same 0 to 1 scale, against 0.97 and 0.99).

With more answer levels the gap is wider. So the model gets two of the three numbers almost for free and works much harder for the third.

Why sharpness is the hard one

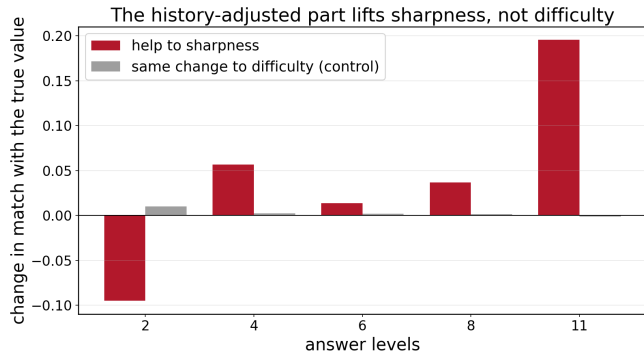


A model can only learn a number as well as its prediction leans on that number.

The prediction leans heavily on score and difficulty. It barely leans on sharpness. Sharpness only changes the prediction for learners whose strength is close to a question's difficulty. For everyone far above or far below, the answer is nearly settled either way, so sharpness makes almost no difference. Add more answer levels and the prediction leans on it even less.

That weak signal is the main reason sharpness is slow and unreliable.

A fix aimed straight at the hard number



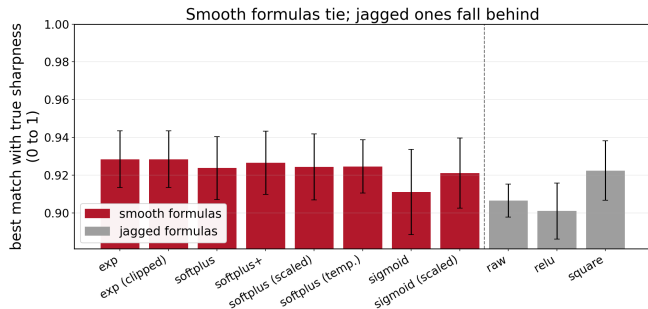
Give each question's sharpness two parts, a fixed part for the question plus a small extra part that shifts with the learner's recent answers.

The match to true sharpness improves, mostly by getting there faster and more reliably, and the gain grows with more answer levels, up to about +0.20 at eleven.

Two controls make it real. The same change to difficulty does almost nothing (grey bars), and a size-matched plain model does not close the gap.

Caveat, at two answer levels it slightly hurts, so this helps richer grading scales, not the simplest right or wrong case.

A tempting explanation we tested and ruled out



Sharpness must be a positive number. A common trick is to learn a free number and pass it through an always-positive formula. We suspected one formula, the exponential, was the secret.

Compared fairly, each given the same starting point and its own best settings, they all tie. Only the crude options lag, the one that can still slip negative and the one that flattens to zero and stops responding.

A follow-up idea, that the exponential helps by changing the training step sizes, also failed its own test. Being positive and smooth helps, the exponential does not. What actually helps is the previous slide's fix, giving sharpness its own dedicated part of the model.

How far to trust the history part

We planted a real, known link between a question's sharpness and the learner's strength, then asked the history-adjusted part to find it.

- It picks up the **direction** of the link.
- It badly **underestimates the size** of the link.
- It shows a little signal even when we planted none, so a small adjustment can be noise, not a real link.

So we read the history part for direction, not for amount. A bigger history adjustment is not the same as a stronger real effect.

Takeaway

One line

A model trained only to predict answers learns each question number only as well as the prediction leans on it.

- Sharpness is the number the prediction barely leans on, so it is learned slowly and only roughly.
- Giving sharpness a fixed part plus a history-adjusted part helps, more so with more answer levels. It is a gain in speed and reliability, and it recovers the order of sharpness, not its exact size.
- We ruled out the idea that a special positive formula was the key. Positivity and smoothness matter, the formula does not.

Honest limits, made-up data, a few repeats, one model type. Next step, a check on real learner data.